Author Statements

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CRediT Author Statement

Role	S. B. Mahafuj Bondhon
Conceptualization	✓
Methodology	✓
Software	✓
Validation	✓
Formal analysis	✓
Investigation	✓
Data curation	✓
Writing – original draft	✓
Visualization	✓

Conflict of Interest Statement

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding Statement

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data Availability Statement

The primary dataset used in this study is the publicly available BubbleML Pool-Boiling Subcooled FC-72 dataset (Hassan et al., BubbleML: A Multiphase Multiphysics Dataset and Benchmarks for Machine Learning, NeurIPS 2023 Datasets and Benchmarks Track), available at the official HPCForge/BubbleML repository and its associated data hosting locations. The exact source release and checksums used by this study are documented in `CHECKSUMS.md`; this repository does not redistribute the raw data.

Code, retained machine-readable result artifacts, configurations, and statistical-analysis scripts for this study are designated for release at:

- GitHub repository: https://github.com/RealmeBTI/bubbleml-submission
- Zenodo archival DOI: NOT YET AVAILABLE. It will be added only after a verified GitHub release and Zenodo archival.

The repository includes a README with environment setup instructions, pinned dependency versions, and stored-result reproduction instructions. Raw data, complete checkpoints, and complete cross-condition exports remain external or unavailable as documented in `ARTIFACT_GAPS.md`; this statement does not claim a full retraining release.

Ethics Statement

This research uses publicly available simulation data and does not involve human participants, human data, or animals; no ethical approval was required.